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Exercise list: *L07 — Class 7: GLS, AR and Student-t errors, hierarchical models, Tobit and probit (book exercises)*

Instructions. Every question is a textbook exercise listed under *Practice* for Class 7 in `Map/class-plan.md`, reproduced verbatim from the book PDF with its source line; blocks follow the books in the order the class plan cites them. Text marked "Added for reference (not in the exercise)" is context taken from the cited page of the same book. Computer exercises are solved in Python (`statsmodels`, `linearmodels`, NumPy/SciPy); every number in the answer key is reproduced by `Listas/L07-book-exercises-check.py`. Data: the Wooldridge data sets come from the `wooldridge` Python package; "Birth weight II" is in `Listas/data/babies.data` (provenance in `Listas/data/README.md`). Show all steps.

Block 1 — Wooldridge, *Introductory Econometrics* (6e)

(Ch. 12, 17, 14)

Question 1. Wooldridge 12.C2 — iterated Cochrane–Orcutt and the long-run propensity.

Source: Wooldridge, J. M. (2016), *Introductory Econometrics: A Modern Approach*, 6th ed., Chapter 12, Computer Exercise C2, pp. 397–398, reproduced verbatim.

C2

(i) Using the data in WAGEPRC, estimate the distributed lag model from Problem 5 in Chapter 11. Use regression (12.14) to test for AR(1) serial correlation.

(ii) Reestimate the model using iterated Cochrane-Orcutt estimation. What is your new estimate of the long-run propensity?

(iii) Using iterated CO, find the standard error for the LRP. (This requires you to estimate a modified equation.) Determine whether the estimated LRP is statistically different from one at the 5% level.

Added for reference (not in the exercise). Problem 5 in Chapter 11 (p. 366) regresses *gprice* on *gwage*, $gwage_{-1}$, …, $gwage_{-12}$ and a constant, where "$gprice = \Delta \log(price)$ and $gwage = \Delta \log(wage)$". Regression (12.14), p. 377, is "$\hat{u}_t$ on $\hat{u}_{t-1}$, for all $t = 2, \ldots, n$".

Solution:

Question 2. Wooldridge 12.C10 — static Phillips curve by Prais–Winsten and Cochrane–Orcutt.

Source: Wooldridge, J. M. (2016), *Introductory Econometrics: A Modern Approach*, 6th ed., Chapter 12, Computer Exercise C10, p. 399, reproduced verbatim.

C10 Use the data in PHILLIPS to answer these questions.

(i) Using the entire data set, estimate the static Phillips curve equation $inf_t = \beta_0 + \beta_1 unem_t + u_t$ by OLS and report the results in the usual form.

(ii) Obtain the OLS residuals from part (i), $\hat{u}_t$, and obtain ρ from the regression $\hat{u}_t$ on $\hat{u}_{t-1}$. (It is fine to include an intercept in this regression.) Is there strong evidence of serial correlation?

(iii) Now estimate the static Phillips curve model by iterative Prais-Winsten. Compare the estimate of β_1 with that obtained in Table 12.2. Is there much difference in the estimate when the later years are added?

(iv) Rather than using Prais-Winsten, use iterative Cochrane-Orcutt. How similar are the final estimates of ρ? How similar are the PW and CO estimates of β_1?

Added for reference (not in the exercise). Table 12.2 (p. 386), data through 1996, $n = 49$: Prais–Winsten $\hat{\beta}_1 = -.716$ (.313), intercept 8.296 (2.231), $\hat{\rho} = .781$.

Solution:

Question 3. Wooldridge 17.3 — partial effects in the truncated part of the Tobit model.
Source: Wooldridge, J. M. (2016), *Introductory Econometrics: A Modern Approach*, 6th ed., Chapter 17, Problem 3, p. 559, reproduced verbatim.

3 (Requires calculus)

(i) Suppose in the Tobit model that $x_1 = \log(z_1)$, and this is the only place z_1 appears in $\mathbf{x}$. Show that

$$\frac{\partial \mathrm{E}(y \mid y > 0, \mathbf{x})}{\partial z_1} = (\beta_1/z_1)\{1 - \lambda(\mathbf{x}\boldsymbol{\beta}/\sigma)[\mathbf{x}\boldsymbol{\beta}/\sigma + \lambda(\mathbf{x}\boldsymbol{\beta}/\sigma)]\}, \tag{17.52}$$

where β_1 is the coefficient on $\log(z_1)$.

(ii) If $x_1 = z_1$, and $x_2 = z_1^2$, show that

$$\frac{\partial \mathrm{E}(y \mid y > 0, \mathbf{x})}{\partial z_1} = (\beta_1 + 2\beta_2 z_1)\{1 - \lambda(\mathbf{x}\boldsymbol{\beta}/\sigma)[\mathbf{x}\boldsymbol{\beta}/\sigma + \lambda(\mathbf{x}\boldsymbol{\beta}/\sigma)]\},$$

where β_1 is the coefficient on z_1 and β_2 is the coefficient on z_1^2.

Solution:

Question 4. Wooldridge 17.C2 — probit and logit for loan approval.
Source: Wooldridge, J. M. (2016), *Introductory Econometrics: A Modern Approach*, 6th ed., Chapter 17, Computer Exercise C2, p. 561, reproduced verbatim.

C2 Use the data in LOANAPP for this exercise; see also Computer Exercise C8 in Chapter 7.

(i) Estimate a probit model of *approve* on *white*. Find the estimated probability of loan approval for both whites and nonwhites. How do these compare with the linear probability estimates?

(ii) Now, add the variables *hrat*, *obrat*, *loanprc*, *unem*, *male*, *married*, *dep*, *sch*, *cosign*, *chist*, *pubrec*, *mortlat1*, *mortlat2*, and *vr* to the probit model. Is there statistically significant evidence of discrimination against nonwhites?

(iii) Estimate the model from part (ii) by logit. Compare the coefficient on *white* to the probit estimate.

(iv) Use equation (17.17) to estimate the sizes of the discrimination effects for probit and logit.

Solution:

Question 5. Wooldridge 17.C3 — Tobit model for pension benefits.
Source: Wooldridge, J. M. (2016), *Introductory Econometrics: A Modern Approach*, 6th ed., Chapter 17, Computer Exercise C3, p. 561, reproduced verbatim.

C3 Use the data in FRINGE for this exercise.

(i) For what percentage of the workers in the sample is *pension* equal to zero? What is the range of *pension* for workers with nonzero pension benefits? Why is a Tobit model appropriate for modeling *pension*?

(ii) Estimate a Tobit model explaining *pension* in terms of *exper*, *age*, *tenure*, *educ*, *depends*, *married*, *white*, and *male*. Do whites and males have statistically significant higher expected pension benefits?

(iii) Use the results from part (ii) to estimate the difference in expected pension benefits for a white male and a nonwhite female, both of whom are 35 years old, are single with no dependents, have 16 years of education, and have 10 years of experience.

(iv) Add *union* to the Tobit model and comment on its significance.

(v) Apply the Tobit model from part (iv) but with *peratio*, the pension-earnings ratio, as the dependent variable. (Notice that this is a fraction between zero and one, but, though it often takes on the value zero, it never gets close to being unity. Thus, a Tobit model is fine as an approximation.) Does gender or race have an effect on the pension-earnings ratio?

Solution:

Question 6. Wooldridge 17.C9 — Tobit demand for eco-labeled apples.

Source: Wooldridge, J. M. (2016), *Introductory Econometrics: A Modern Approach*, 6th ed., Chapter 17, Computer Exercise C9, pp. 562–563, reproduced verbatim.

C9 Use the data in APPLE for this exercise. These are telephone survey data attempting to elicit the demand for a (fictional) "ecologically friendly" apple. Each family was (randomly) presented with a set of prices for regular apples and the ecolabeled apples. They were asked how many pounds of each kind of apple they would buy.

(i) Of the 660 families in the sample, how many report wanting none of the ecolabeled apples at the set price?

(ii) Does the variable *ecolbs* seem to have a continuous distribution over strictly positive values? What implications does your answer have for the suitability of a Tobit model for *ecolbs*?

(iii) Estimate a Tobit model for *ecolbs* with *ecoprc*, *regprc*, *faminc*, and *hhsize* as explanatory variables. Which variables are significant at the 1% level?

(iv) Are *faminc* and *hhsize* jointly significant?

(v) Are the signs of the coefficients on the price variables from part (iii) what you expect? Explain.

(vi) Let β_1 be the coefficient on *ecoprc* and let β_2 be the coefficient on *regprc*. Test the hypothesis $\mathrm{H}_0 \colon -\beta_1 = \beta_2$ against the two-sided alternative. Report the p-value of the test. (You might want to refer to Section 4-4 if your regression package does not easily compute such tests.)

(vii) Obtain the estimates of $\mathrm{E}(ecolbs \mid \mathbf{x})$ for all observations in the sample. [See equation (17.25).] Call these $\widehat{ecolbs}_i$. What are the smallest and largest fitted values?

(viii) Compute the squared correlation between $ecolbs_i$ and $\widehat{ecolbs}_i$.

(ix) Now, estimate a linear model for *ecolbs* using the same explanatory variables from part (iii). Why are the OLS estimates so much smaller than the Tobit estimates? In terms of goodness-of-fit, is the Tobit model better than the linear model?

(x) Evaluate the following statement: "Because the R-squared from the Tobit model is so small, the estimated price effects are probably inconsistent."

Solution:

Question 7. Wooldridge 14.C5 — occupation dummies in the fixed effects wage equation.

Source: Wooldridge, J. M. (2016), *Introductory Econometrics: A Modern Approach*, 6th ed., Chapter 14, Computer Exercise C5, p. 454, reproduced verbatim.

C5

(i) In the wage equation in Example 14.4, explain why dummy variables for occupation might be important omitted variables for estimating the union wage premium.

(ii) If every man in the sample stayed in the same occupation from 1981 through 1987, would you need to include the occupation dummies in a fixed effects estimation? Explain.

(iii) Using the data in WAGEPAN include eight of the occupation dummy variables in the equation and estimate the equation using fixed effects. Does the coefficient on *union* change by much? What about its statistical significance?

Added for reference (not in the exercise). Example 14.4 (p. 443) regresses log(*wage*) on *educ*, *black*, *hispan*, *exper*, $exper^2$, *married*, *union* and a full set of year dummies; in the fixed effects column of Table 14.2 only $exper^2$ ($-.0052$), *married* (.047) and *union* (.080, se .019) remain.

Solution:

Block 2 — Greenberg, *Introduction to Bayesian Econometrics* (1e)

(Ch. 4, 8, 9)

Question 8. Greenberg 8.3 — birth weight with Student-t errors; marginal likelihoods.

Source: Greenberg, E. (2007), *Introduction to Bayesian Econometrics*, 1st ed., Chapter 8, Exercise 8.3, p. 132, reproduced verbatim (Exercise 8.1, to which it refers, reproduced verbatim from the same page).

8.3 Repeat Exercise 8.1 for errors distributed as Student-t with five degrees of freedom. Compute marginal likelihoods and compare the models.

8.1 Download the data in "Birth weight II" from the StatLabs Web site, `www.stat.berkeley.edu/users/statlabs/labs.html`. The variables are birth weight (ounces, 999 unknown), gestation (length of pregnancy in days, 999 unknown), parity (0 = firstborn, 9 is unknown), age (mother's age in years), height (mother's height in inches), weight (mother's prepregnancy weight in pounds), smoke (0 not now, 1 yes now, 9 = unknown).

Specify a linear regression, where the response variable is birth weight, and the covariates are a constant and the remaining variables in the data set. Eliminate observations for which you do not have complete information on the variables. Specify prior distributions for the coefficients and the variance (or precision). Summarize the results for the dummy variable for smoking. Does smoking by the mother appear to affect the baby's birth weight?

Solution:

Question 9. Greenberg 8.5 — Gibbs algorithm for regression with Student-t errors.

Source: Greenberg, E. (2007), *Introduction to Bayesian Econometrics*, 1st ed., Chapter 8, Exercise 8.5, p. 132, reproduced verbatim.

8.5 Write out an algorithm to sample for β, σ^2, and λ for the linear regression model with Student-t errors of Section 8.1.2.

Solution:

Question 10. Greenberg 4.6 — i.i.d. implies exchangeable.

Source: Greenberg, E. (2007), *Introduction to Bayesian Econometrics*, 1st ed., Chapter 4, Exercise 4.6, p. 59, reproduced verbatim.

4.6 Show that i.i.d. variables are exchangeable.

Solution:

Question 11. Greenberg 9.2 — binary panel data (random-effects probit).

Source: Greenberg, E. (2007), *Introduction to Bayesian Econometrics*, 1st ed., Chapter 9, Exercise 9.2, p. 151, reproduced verbatim.

9.2 Explain how to analyze a panel data set where the response variable is binary, that is, $y_{it} \in \{0, 1\}$, the latent data are determined by $y^*_{it} = x'_{it}\beta + w'_{it}b_i + u_{it}$, and the remainder of the specification is given in (9.11). (Note: Chib and Carlin (1999) provide several algorithms for this model.)

Added for reference (not in the exercise). Equation (9.11), p. 146: $y_i = X_i\beta + W_i b_i + u_i$, $u_i \,|\, h_u \sim N_T(0, h_u^{-1} I_T)$, $\beta \sim N_{K_1}(\beta_0, B_0)$, $h_u \sim G(\alpha_0/2, \delta_0/2)$, $b_i \,|\, D \sim N_{K_2}(0, D)$, $D^{-1} \sim W_{K_2}(\nu_0, D_0)$.

Solution:

Question 12. Greenberg 9.3 — panel model with Student-t errors and random effects.

Source: Greenberg, E. (2007), *Introduction to Bayesian Econometrics*, 1st ed., Chapter 9, Exercise 9.3, pp. 151–152, reproduced verbatim.

9.3 The basic panel data model of Equation (9.11) can be modified to specify Student-t errors for either or both of u_i and b_i. To do so for both, make the following changes and additions to those equations:

$$\begin{aligned} u_i \,|\, h_u, \lambda_i &\sim N_T(0, (\lambda_i h_u)^{-1} I_T), \\ \lambda_i &\sim G(\nu_l/2, \nu_l/2), \\ b_i \,|\, D, \eta_i &\sim N_{K_2}(0, \eta_i^{-1} D), \\ \eta_i &\sim G(\nu_h/2, \nu_h/2). \end{aligned}$$

Show that, in the modified model,

$$u_i \,|\, h_u \sim t_T(\nu_l, 0, I_T) \quad \text{and} \quad b_i \,|\, D \sim t_{K_2}(\nu_h, 0, D).$$

Develop an algorithm to sample from $\pi(\beta, h_u, D \,|\, y)$.

Solution:

Question 13. Greenberg 8.6 — Gibbs algorithm for the Tobit model with Student-t errors.
Source: Greenberg, E. (2007), *Introduction to Bayesian Econometrics*, 1st ed., Chapter 8, Exercise 8.6, p. 132, reproduced verbatim.

8.6 Write out an algorithm to sample for β, σ^2, y^*_{Ci}, and λ for the Tobit model of Section 8.2.1 with Student-t errors.

Solution:

Question 14. Greenberg 8.7 — Gibbs algorithm for binary data with Student-t errors.
Source: Greenberg, E. (2007), *Introduction to Bayesian Econometrics*, 1st ed., Chapter 8, Exercise 8.7, p. 132, reproduced verbatim.

8.7 Write out an algorithm to sample for β, y^*, and λ for the binary data model with Student-t errors of Section 8.2.2.

Solution:

Block 3 — Morettin & Singer, *Estatística e Ciência de Dados* (2021)

(Ch. 6)

Question 15. Morettin & Singer 6.18 — odds ratios in a logistic regression (statement in Portuguese).
Source: Morettin, P. A. and Singer, J. M. (2021), *Estatística e Ciência de Dados* (versão preliminar, abril 2021), Capítulo 6, Exercício 18, pp. 241–242, reproduced verbatim.

18) Para estudar a associação entre gênero (1=Masc, 0=Fem) e idade (anos) e a preferência (1=sim, 0=não) pelo refrigerante Kcola, o seguinte modelo de regressão logística foi ajustado aos dados de 50 crianças escolhidas ao acaso:

$$\log\left\{\frac{\pi_i(x_i, w_i)}{1-\pi_i(x_i, w_i)}\right\} = \alpha + \beta x_i + \gamma(w_i - 5),$$

em que x_i (w_i) representa o gênero (idade) da i-ésima criança e $\pi_i(x_i, w_i)$ a probabilidade de uma criança do gênero x_i e idade w_i preferir Kcola. As seguintes estimativas para os parâmetros foram obtidas:

Parâmetro	Estimativa	Erro padrão	Valor p
α	0,69	0,12	$< 0{,}01$
β	0,33	0,10	$< 0{,}01$
γ	-0,03	0,005	$< 0{,}01$

a) Interprete os parâmetros do modelo por intermédio de chances e razões de chances.

b) Com as informações acima, estime a razão de chances de preferência por Kcola correspondente à comparação de crianças do mesmo gênero com 10 e 15 anos.

c) Construa intervalos de confiança (com coeficiente de confiança aproximado de 95%) para $\exp(\beta)$ e $\exp(\gamma)$ e traduza o resultado em linguagem não técnica.

d) Estime a probabilidade de meninos com 15 anos preferirem Kcola.

Solution:

Question 16. Morettin & Singer 6.21 — diabetes duration and retinopathy (statement in Portuguese).

Source: Morettin, P. A. and Singer, J. M. (2021), *Estatística e Ciência de Dados* (versão preliminar, abril 2021), Capítulo 6, Exercício 21 and Tabela 6.17, p. 242, reproduced verbatim.

21) Os dados da Tabela 6.17 contem dados de uma investigação cujo objetivo era estudar a relação entre a duração de diabete e a ocorrência de retinopatia (uma moléstia dos olhos). Ajuste um modelo de regressão logística para avaliar a intensidade dessa relação.

Sugestão: Considere o ponto médio de cada intervalo como valor da variável explicativa.

Tabela 6.17: Frequências de retinopatia

Duração da	Retinopatia	
Diabete (anos)	Sim	Não
0 - 2	17	215
3 - 5	26	218
6 - 8	39	137
9 - 11	27	62
12 - 14	35	36
15 - 17	37	16
18 - 20	26	13
21 - 23	23	15

Solution: